

Smart Farm Assistant - Detailed ML & Dataset Report

A detailed point-wise breakdown of the exact models and datasets used in this project.

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1. Crop Recommendation Model

*** Current Project Status: Fully Implemented (API Endpoint: /api/crop/predict)**

- Purpose: Suggests the optimal crop to maximize yield for farmers.
- *Input Features Used in Frontend:*
 - Soil Type, Nitrogen (N), Phosphorus (P), Potassium (K)
 - Temperature, Humidity, Rainfall, pH levels
- Output: Predicted crop name and calculation confidence percentage.
- *Associated Dataset Used in Admin Portal:*
 - Name: 'Crop Recommendation Training Data'
 - Format: Tabular (CSV)
 - Size: 45 MB (~125,000 records)
 - Accuracy: 94.2% (Active Status)
- Algorithm Architecture: Modeled for Ensemble Learning (e.g., Random Forest Classifier).

2. Rain & Weather Forecasting Model

*** Current Project Status: Implemented (API Endpoint: /api/rain/predict)**

- Purpose: Predicts probability of rain for a specific geographical location.
- *Input Features Used in Frontend:*
 - User Location string input
- Output: Rain probability percentage.
- *Associated Dataset Used in Admin Portal:*
 - Name: 'Regional Weather Logs 2025'
 - Format: JSON
 - Size: 18 MB (8,760 hourly records)
 - Accuracy: N/A (Archived Status)
- Algorithm Architecture: Modeled for Time-series logic.

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3. Plant Leaf Disease Detection Model

*** Current Project Status: UI Interactive (File Upload active, ML bridging pending)**

- Purpose: Visually analyzes crop leaf photos to identify specific crop diseases.

- *Input Features Used in Frontend:*

- Drag-and-drop Image Upload (JPEG/PNG).

- *Associated Dataset Used in Admin Portal:*

- Name: 'Leaf Disease Image Collection V2'

- Format: Images (ZIP Archive)

- Size: 1.2 GB (15,420 annotated images)

- Accuracy: 91.8% (Active Status)

- Algorithm Architecture: Convolutional Neural Network (CNN) image processing.

4. Market Price Forecasting Model

*** Current Project Status: UI Dashboard Implemented**

- Purpose: Streams real-time commodity prices to inform selling windows.

- *Input Features Used in Frontend:*

- Real-time chart visualization.

- *Associated Dataset Used in Admin Portal:*

- Name: 'Market Prices Real-time Sync'

- Format: API Stream

- Size: Continuous (450,000+ records processed)

- Accuracy: 98.5% (Active Status)